

M1 Carbine

PAGES FROM THE ORIGINAL MANUAL

FM 23-7, Basic Field Manual – U.S. Carbine, Caliber .30, M1
(War Department, 1942)

WHAT'S INSIDE

Chapter 1, Mechanical Training: description, taking it apart and putting it back together, cleaning, how it works, operation, stoppages, ammunition and safety.

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FM 23-7

WAR DEPARTMENT

BASIC FIELD MANUAL

**U. S. CARBINE, CALIBER
.30, M1**

May 20, 1942

BASIC FIELD MANUAL

U. S. CARBINE, CALIBER .30 M1

CHAPTER I

MECHANICAL TRAINING

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SECTION I

GENERAL

■ 1. DESCRIPTION OF CARBINE.—The U. S. carbine, caliber .30, M1, is a self-loading shoulder weapon (see fig. 1). It is gas operated, magazine fed, and air cooled. It weighs approximately 5 pounds. The weapon is fed from a box type magazine having a capacity of 15 rounds.

■ 2. GENERAL DATA.—*a. Dimensions.*—(1) *Barrel.*

Diameter of bore.....inches... .30

Number of grooves.....4

Length of barrel.....inches... 17.75

(2) *Carbine.*

Over-all length of carbine.....inches... 35.50

Sight radius.....do... 22.

b. Weight.

Carbine without sling.....pounds... 5.12

Carbine with sling.....do... 5.23

Magazine, empty.....do... .17

Magazine, loaded.....do... .57

Total weight w/sling and loaded
magazine.....do... 5.80

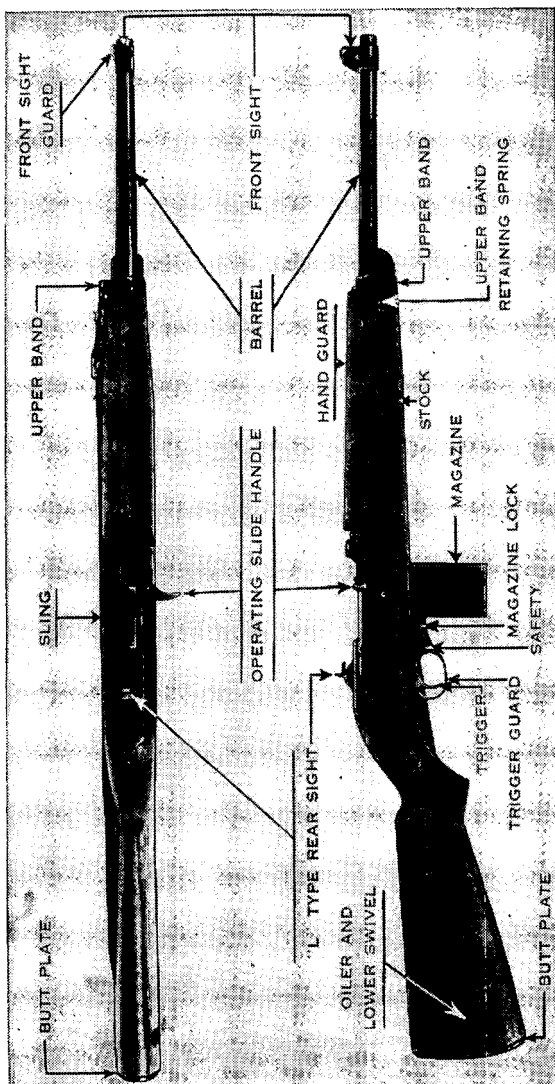


FIGURE 1.—U. S. carbine, caliber .30, M1.

■ 3. MISCELLANEOUS DATA.

Muzzle velocity-----feet per second-- 1900

Pressure in chamber (approx.)

pounds per square inch-- 40,000

Weight of ball cartridge (approx.)

grains-- 195

Weight of bullet (approx.)-----grains-- 110

■ 4. REAR SIGHT.—*a.* Models of the carbine initially issued will be equipped with an L-type rear sight, consisting of

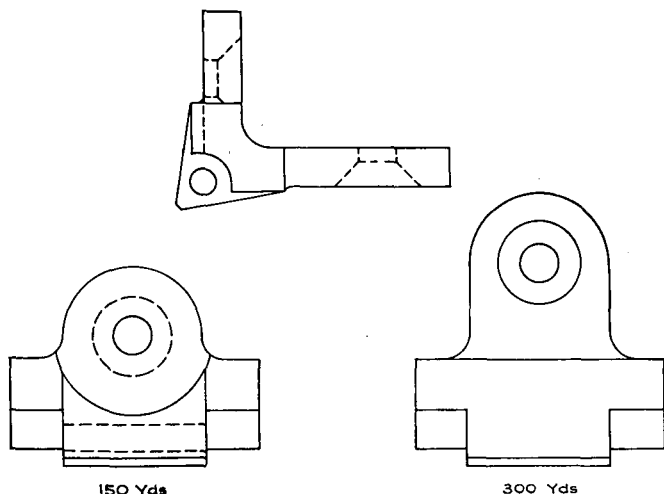


FIGURE 2.—L-type rear sight.

two arms at right angles, each pierced with an aperture. A flat spring is placed between the sight leaf and sight base to retain the sight leaf in position. Either sight may be raised into position by turning with the fingers, and the leaf is held correctly by the pressure of the spring. The apertures provided are computed for ranges of 150 and 300 yards. With this sight it will be necessary to aim off the target to secure intermediate changes in range and windage corrections. (See fig. 2.)

b. Later models of the carbine will be equipped with an adjustable sight generally similar to that shown in other figures in this manual. This sight will be graduated from 100 to 300 yards in increments of 50 yards and will provide for 3 points of left and 3 points of right windage. The amount which changes of elevation and windage will move the point of strike must be determined by experiment.

■ 5. FRONT SIGHT.—The front sight is the post type, protected by wings and adjusted laterally during assembly at the arsenal. It is locked in position, after adjustment, by riveting part of the metal base into the lock seat with a punch.

■ 6. NOMENCLATURE AND REFERENCES.—*a. Nomenclature.*—The soldier should be familiar with the names of those parts of the carbine which are frequently referred to in drill and range practice (figs. 3 to 6, inclusive).

b. References.—Safety precautions to be observed by troops are complete in this manual. Range officers, the officer in charge of firing, and the commander responsible for the location of ranges and conduct of firing should refer to AR 750-10 for additional safety precautions.

SECTION II

DISASSEMBLY AND ASSEMBLY

■ 7. WHEN TAKEN UP.—Training in disassembly and assembly is taken up as soon as practicable after the soldier receives his carbine. This training is completed before the individual does any firing with the weapon.

■ 8. ORGANIZATION.—In the company or platoon, men armed with the carbine are organized into one or more groups under their officers or selected noncommissioned officers as instructors and supervisors.

■ 9. CARE TO BE EXERCISED.—*a.* The carbine can be readily disassembled and assembled without applying force. The application of force is prohibited.

b. The weapon will not be disassembled or assembled against time. In all practice in disassembling the carbine, individuals will be taught to lay the parts out on a smooth, clean surface in the proper sequence for assembling.

■ 10. DISASSEMBLING.—*a. General.*—Disassembly of the carbine by the soldier is limited to those steps required for proper care and maintenance of the weapon. Further disassembly will not be made except under the supervision of an officer or ordnance personnel. Only the following parts may be removed by the individual soldier without supervision:

Sling and oiler (and lower sling swivel).

Magazine.

Hand guard.

Stock assembly.

Operating slide spring.

Operating slide spring guide rod.

Guard assembling pin.

Trigger group assembly.

Operating slide.

Bolt assembly.

Barrel and receiver assembly.

b. Sequence.—The disassembly of the carbine authorized to be performed by the soldier without supervision is made in the following sequence:

(1) *Sling and oiler (and lower sling swivel).*—Unsnap and remove sling from the upper sling swivel. Remove oiler and attached sling from the recess in the butt of the stock.

(2) *Magazine.*—Hold the carbine, muzzle to the front, between the right side of the body and right forearm. Hold the magazine with the left hand. With the thumb of the right hand, press the magazine lock from the right side to the left and withdraw the magazine downward out of the receiver. *CAUTION: Do not let the magazine drop to the ground.*

(3) *Hand guard.*—Place the carbine on a level surface, resting the muzzle so that the head of the upper band screw is up and to the left. Using the rim of a carbine cartridge, loosen the upper band screw about $\frac{1}{8}$ inch. Turn the barrel over. Depress the upper band retaining spring with the base of the cartridge, and slide the upper band over the retaining spring and off the stock. With the left hand, slide the hand guard forward until its rear end is disengaged from the groove in the front end of the receiver, and remove the hand guard from the barrel.

(4) *Stock assembly.*—Grasp the small of the stock with the right hand and the barrel with the left hand, palm up. Push the safety to the left. Raise the muzzle end of the barrel from the stock until the receiver is released from the stock abutment. Remove the stock from the barrel, trigger group assembly, and receiver.

(5) *Operating slide spring and guide rod.*—Place the barrel and attached assemblies on a level surface, muzzle to



FIGURE 3.—Three main groups, U. S. carbine, caliber .30, M1.

the left, operating slide handle up. Grasp the operating spring guide rod between the thumb and forefinger of the right hand and pull the rod to the rear, disengaging it from its seat in the operating slide. Raise the rod slightly and withdraw the operating spring to the left from its well in the receiver (see fig. 7). Separate the spring from its guide rod.

(6) *Trigger group assembly.*—Cock the hammer by pulling the operating slide to the rear and pushing it forward. Turn

the barrel so that the operating slide handle is down. Punch the guard assembling pin from its seat by starting it with the small end of the operating spring guide rod held in the right hand. Pull out pin from far side with left hand.

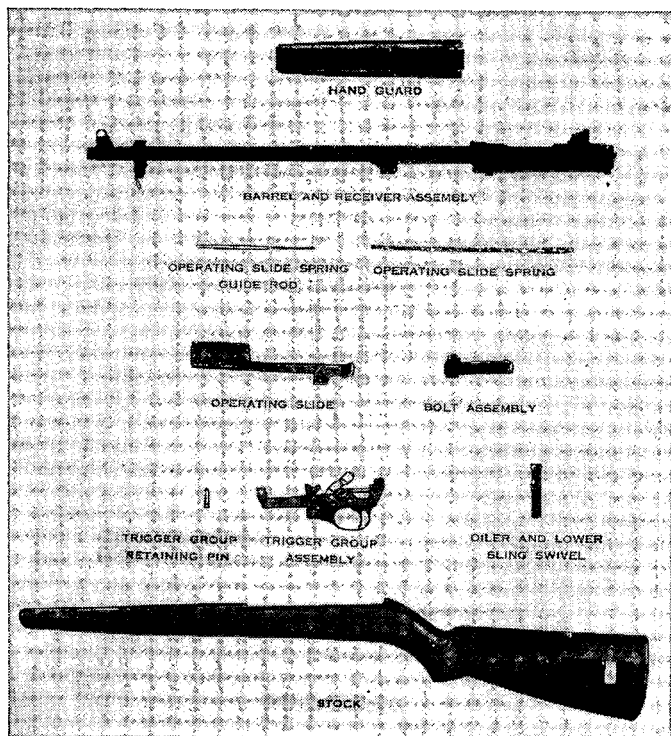


FIGURE 4.—Groups and parts to be removed for care and cleaning.

Grasp the barrel with the left hand and the trigger group assembly with the right hand as shown in figure 8. The trigger group assembly may now be removed by sliding it to the left and disengaging its undercut grooves from the corresponding grooves in the receiver.

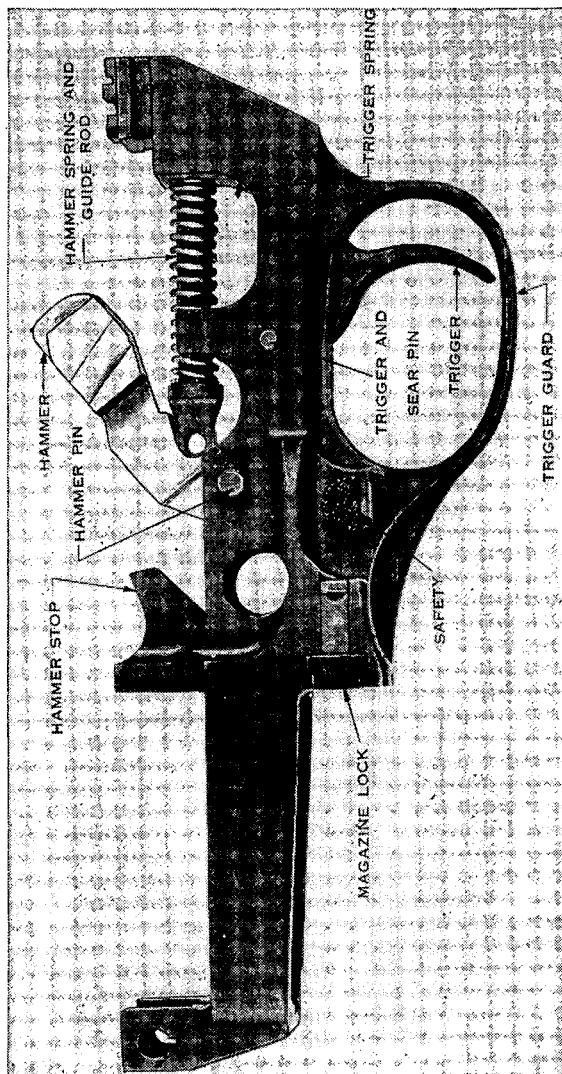


FIGURE 5.—Trigger group assembly with hammer in cocked position.



FIGURE 6.—Trigger group assembly with hammer in firing position.

(7) *Operating slide* (see fig. 9).—Rest the barrel on a level surface, muzzle to the front, receiver down. Hold the rear end of the receiver against the body with the left hand, thumb resting on top of the left side of the operating slide. Grasp the operating slide handle with the thumb and forefinger of the right hand and draw the slide slowly to the rear until the lug on the inner side of the rear end of the slide is opposite

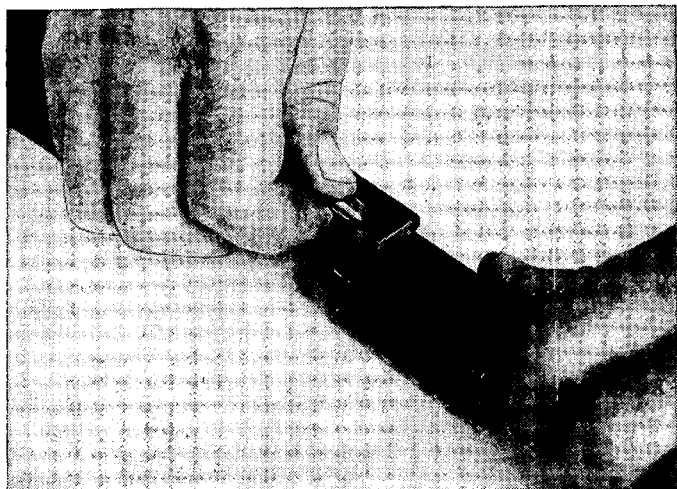


FIGURE 7.—Removing operating slide spring and guide rod.

the operating slide dismounting notch in the operating groove of the receiver. Pull the operating handle to the right and up and disengage the rear end of the slide from the operating lug on the bolt. The slide may now be removed from the barrel by a downward pressure of the left thumb on the left side of the slide, thus disengaging the lug on the slide from the left guiding groove in the under side of the barrel.

(8) *Bolt assembly*.—Hold the barrel as described in (7) above. Grasp the operating lug on the bolt between the thumb and forefinger of the right hand and move the bolt so that its front end is just in rear of the locking recesses in the receiver. Raise the front end of the bolt about 1 inch.

Rotate the bolt slightly to the right so that the tail of the firing pin is opposite its notch in the bottom of the receiver. Lift the bolt from the receiver.

■ 11. ASSEMBLING.—The carbine and its component groups are assembled in the reverse order of their disassembly.

a. *Bolt assembly.*—Grasp the operating lug of the bolt

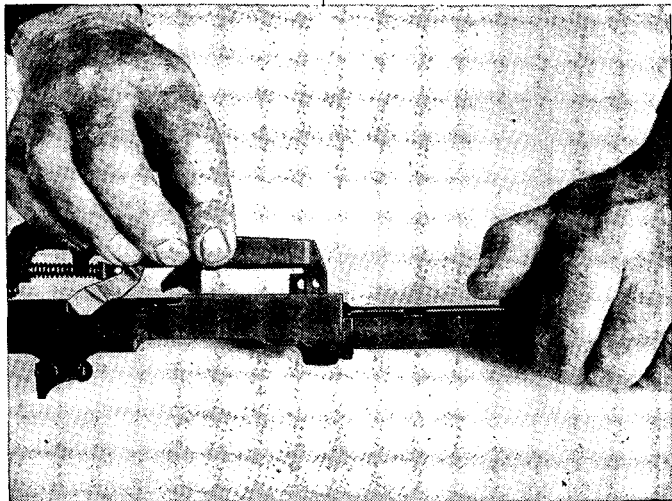


FIGURE 8.—Sliding trigger group assembly off receiver.

between the thumb and forefinger of the right hand. Hold the bolt so that the tail of the firing pin is opposite its notch in receiver, then lower the bolt into position.

b. *Operating slide.*—Grasp the forward end of the slide in the right hand, palm up. Slide the bolt forward until its forward end is about $1\frac{1}{2}$ inches from the chamber and hold it in this position with the left thumb. Engage the operating lug of the bolt in the operating cam groove of the slide. Raise the forward end of the slide so that the dismounting lug on its left side is opposite the notch in the left groove on the under side of the barrel. Then, by slightly twisting the slide to the right, engage the operating lugs of the slide in the

operating grooves of the barrel. Move the slide and bolt to the rear until the operating slide lug is seated in its groove in the receiver. Close the bolt. (See fig. 10.)

c. Trigger group assembly.—Place the barrel and receiver on its left side, muzzle to the left. Replace the trigger group assembly by engaging its undercut grooves with the cor-

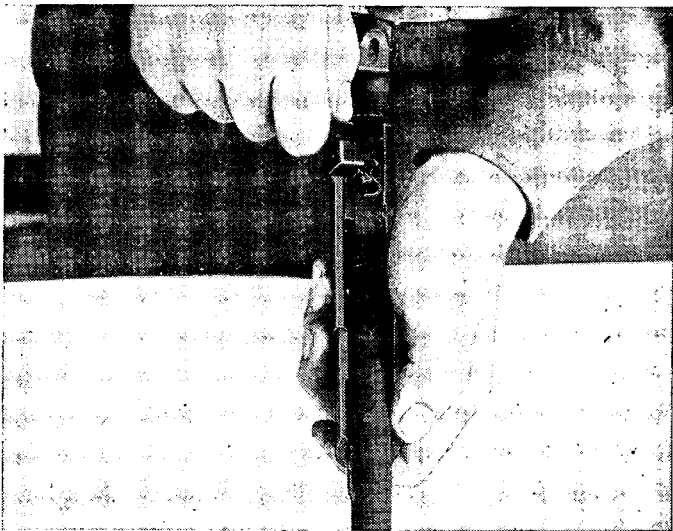


FIGURE 9.—Removing the operating slide.

responding grooves in the receiver. Replace the guard assembling pin.

d. Operating slide spring and guide rod.—Insert the small end of the guide rod in the loosely coiled end of the spring. Insert the closely coiled end of the spring into its well in the receiver. Steady the barrel with the left hand, and with the thumb and forefinger of the right hand on the shoulder of the guide rod, compress the spring so that the end of the guide rod may be inserted in its seat in the slide.

e. Stock assembly and hand guard.—Make certain the safety is to the left. Replace the barrel and assembled

groups in the stock. Replace the hand guard. Slide the upper band down over the end of the hand guard and stock until it is engaged by the upper band retaining spring. Tighten the upper band screw.

f. Sling and oiler (and lower sling swivel).—Attach lower loop of the sling to the oiler. Replace the oiler in its recess in the stock. Attach upper end of the sling to the upper sling swivel.

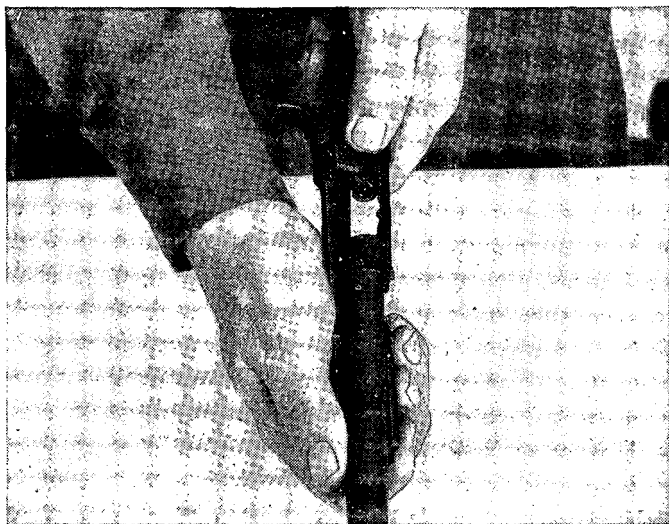


FIGURE 10.—Assembling operating slide to receiver.

SECTION III

CARE AND CLEANING

■ 12. GENERAL.—The attention given to a weapon of this type determines largely whether it will function properly and shoot accurately. The bore and chamber must be kept in good condition for accurate shooting. It is essential that the receiver and moving parts be kept clean, lubricated, and in good condition for reliability in functioning. Magazines must

be kept free from rust, grit, gum, and dents or other damage in order to function properly.

■ 13. IN GARRISON AND CAMP.—*a. General.*—Keep the carbine well cleaned and oiled so as to preserve its condition and appearance during the periods when no firing is being done. Carbines in the hands of troops should be inspected daily to insure proper condition and cleanliness.

b. Bore.—The bore of the carbine will always be cleaned with a cleaning rod inserted from the muzzle. The rod used should be of such length as to permit cleaning the bore without damage to the face of the bolt. If a longer rod is used the bolt must be removed or the face of the bolt protected. This can be done in several ways. The simplest way is to cover the face of the bolt with a cloth stuffed into the receiver. To clean the bore, attach a cloth patch to the cleaning rod. Insert the cleaning rod into the bore at the muzzle and move it back and forth several times. Remove the patch. *CAUTION: In cleaning the bore, care must be taken not to foul the gas port with threads from the cleaning patch itself or with other extraneous matter.* Repeat until several successive patches come out absolutely clean. Inspect the bore. If the bore is clean, saturate a patch in light rust-preventive compound and swab the bore thoroughly so that all interior surfaces have a coating of the compound. If the bore still has residue in it swab it with a patch saturated with rifle bore cleaner, until it is clean, then apply the rust preventive.

c. Chamber.—The chamber should be cleaned when the bore is cleaned. A roughened or rusty chamber may cause cartridges to stick. To clean the chamber use a chamber cleaning brush and scrub the chamber vigorously. After this scrubbing, the chamber is further cleaned and oiled in the process of cleaning the bore.

d. Exterior surfaces.—To clean the screw heads and crevices, use a small cleaning brush or small stick. To clean the metal surfaces, wipe with a dry cloth to remove moisture, perspiration, and dirt; then wipe with a cloth containing a small quantity of light preservative lubricating oil. A light film of this oil should be applied to all moving parts, and this protective film should be maintained at all times. The

stock and hand guard should be wiped with raw linseed oil.

e. Magazines.—It is imperative that magazines be given the best of care and kept in perfect condition. They should be disassembled, wiped clean and dry, and thinly coated with oil. Dirt that gets into them must be removed. In handling magazines, care should be taken to avoid denting or bending them, especially the lips of the mouth of the magazine.

f. After cleaning.—After cleaning the carbine, place it in the rack without covering and without a plug in the muzzle or chamber. Muzzle covers, carbine covers, rack covers, and plugs must not be used because they collect moisture and promote rust. To protect the carbines from dust, covers may be placed over racks when squad rooms or tents are swept; they must be removed after the rooms have been swept.

■ 14. PREPARATORY TO FIRING.—The following procedure before firing insures efficient functioning of the carbine:

- a.* Dismount main groups.
- b.* Wipe oil or grease from the bore.
- c.* Thoroughly clean and lightly oil all metal parts. Use light preservative lubricating oil.
- d.* Apply a thin, uniform coating of light preservative lubricating oil to the parts listed below:
 - (1) Bolt lugs (locking and operating).
 - (2) Bolt guides.
 - (3) Cocking cam on bolt.
 - (4) Piston.
 - (5) Contact surfaces of barrel and operating slide.
 - (6) Operating slide cam.
 - (7) Operating slide guide groove in receiver and barrel.
 - (8) Operating slide spring.
 - (9) Operating slide spring guide rod.

CAUTION: Do not apply oil to the under surface of the bolt, as the introduction of oil into the chamber may lead to the generation of excessive pressure.

e. Assemble carbine and rub all outer surfaces with a lightly oiled rag to remove dust.

■ 15. AFTER FIRING.—The bores of all carbines must be thoroughly cleaned by the evening of the day on which they are

fired. They should be cleaned in the same manner for the next three days. *CAUTION: Under no circumstances will metal fouling solution be used in the carbine.*

a. Cleaning immediately after firing, or as soon as possible.—For this purpose water must be used; warm water is good, but warm, soapy water is better. Hold the carbine bottom side up, so that no water will enter the gas port. Run several wet patches through the bore. Remove the patch section from the cleaning rod, substitute the brush, and work this back and forth through the bore several times. Care should be used to see that the brush goes all the way through the bore before the direction is reversed. Detach the brush and run several wet patches through the bore, removing them from the breech end. Follow this with dry patches until the patches come out clean and dry. Saturate a patch in light preservative lubricating oil and push it through the bore, holding the rifle, top side up, so that some of the oil will flow into the gas port. *CAUTION: In cleaning the bore, be careful not to foul the cleaning patch in the gas port.*

b. Complete cleaning.—This cleaning should be done as soon as possible after that described in *a* above. If the carbine is to be fired the next day proceed as in paragraph 14. If the carbine is not to be fired in the next few days repeat procedure in *a* above for 3 days. In addition, the instructions prescribed in paragraphs 13 and 14 will be observed.

■ 16. ON RANGE OR IN FIELD.—The carbine must be kept clean and free from dirt and properly lubricated with oil. To obtain its maximum efficiency the following points must be observed:

a. Never fire a carbine with dust, dirt, mud, or snow in the bore.

b. Keep the chamber clean and free from oil and dirt.

c. Never leave a patch, plug, or other obstruction in the chamber or bore. Neglect of this precaution may result in serious injury.

d. If the carbine gives indications of lack of lubrication and excessive friction, apply additional oil to the parts. Excessive friction exists if the empty cases are being ejected to the right rear. When this occurs, oil should be applied at the first

opportunity, as failures to feed and eject will occur if the condition is not corrected.

e. Keep a light coating of oil on all other metal parts.

f. Oil the piston without loosening the gas cylinder nut.

g. In general, it should not be necessary to remove any of the parts of the carbine in the field for cleaning except the trigger group assembly, the operating slide spring and rod, the operating slide and the bolt assembly.

h. During range firing a well-qualified man should be placed in charge of the cleaning of carbines at the cleaning racks or tables.

■ 17. PREPARATION FOR STORAGE.—*a.* Light preservative lubricating oil is the most suitable oil for preserving the mechanism of carbines. This oil is efficient for preserving the polished surfaces, the bore, and the chamber for a period of from 2 to 6 weeks, depending on the climatic and storage conditions.

b. Light rust-preventive compound is efficient for preserving the polished surfaces, the bore, and the chamber for a period of 1 year or less, depending on the climatic and storage conditions.

c. The carbine should be cleaned and prepared with particular care. The bore, all parts of the mechanism, and the exterior of the carbine should be thoroughly cleaned and then dried completely with rags. In damp climates particular care must be taken to see that the rags are dry. After drying a metal part it should not be touched with bare hands. All metal parts should then be coated either with light preservative lubricating oil or light rust-preventive compound, depending on the length of storage. (See *a* and *b* above.) Application of the rust-preventive compound to the bore of the carbine is best done by dipping the cleaning brush in the compound and running it through the bore two or three times. Before placing the carbine in the packing chest see that the bolt is in its forward position and that the firing pin is released. Paint the wooden supports at the butt and muzzle with rust-preventive compound. Then, handling the carbine by the stock and hand guard only, it should be placed in the packing chest. Under no circumstances should a carbine be placed in storage in a cloth or other cover or

with a plug in the bore. Such articles collect moisture which causes the weapon to rust.

■ 18. **CLEANING OF RIFLES AS RECEIVED FROM STORAGE.**—*a.* Car-bines which have been stored in accordance with paragraph 17 will be coated with either light preservative lubricating oil or light rust-preventive compound. Car-bines received from ordnance storage will, in general, be coated with heavy rust-preventive compound. Use a dry-cleaning solvent to remove all traces of the compound or oil, particular care being taken that all recesses in which springs or plungers operate are cleaned thoroughly. After using the dry-cleaning solvent make sure it is completely removed from all parts. Then follow instructions contained in paragraph 13. If the car-bines are to be fired immediately, follow instructions contained in paragraph 14.

b. Dry-cleaning solvent is a petroleum distillate used for removing grease. It is noncorrosive and of low inflammability. It is generally applied with rag swabs to large parts and as a bath for small parts. The surfaces must be thoroughly dried immediately after removal of the solvent. To avoid leaving finger marks, which are ordinarily acid and induce corrosion, gloves should be worn by persons handling parts after this cleaning. Dry-cleaning solvent will attack and discolor rubber.

SECTION IV

FUNCTIONING

■ 19. **WHEN TAKEN UP.**—Instruction in functioning will be taken up after instruction in the disassembly, assembly, and care and cleaning of the carbine.

■ 20. **DESCRIPTION OF CYCLE.**—*a. Rearward movement.*—(1) When the carbine is loaded and the bolt is closed, the hammer is held in the cocked position by the front end of the sear engaging in the sear notch on the hammer. The trigger lip under action of the trigger spring is seated under the rear end of the sear. If pressure is then applied to the trigger, the trigger pivots about the trigger pin causing the trigger lip to rise and carry with it the rear end of the sear. The sear also pivots about the trigger pin and now has its

front end drawn downward to the point where it disengages from the sear notch on the hammer. The hammer, thus released, is actuated by the hammer spring which causes it to rotate about the hammer pin and strike the firing pin, through which its blow is transmitted to the primer of the cartridge. The above action can take place only if the tang of the firing pin is seated in the receiver. This cannot occur unless the bolt is fully rotated into its locked position. If the hammer should strike the firing pin when the bolt is in any other position the cartridge will not fire; the effect will be to deaden the hammer blow and rotate the bolt toward its locked position. In addition, the safety must be in its off position so that it will not lock the trigger and thus prevent the release of the hammer.

(2) When the bullet passes the gas port some of the powder gases pass through the port in the barrel into the cylinder where they strike the piston and then escape through the cylinder port. The piston is driven sharply to the rear about $\frac{1}{4}$ inch, where its motion is stopped by the gas cylinder nut. The sharp rearward motion of the piston is transmitted by contact to the operating slide, driving it to the rear. The rearward movement of the slide compresses the operating spring.

(3) The initial movement of the operating slide to the rear for about $\frac{3}{8}$ inch is independent of the bolt mechanism, the operating lug on the bolt merely sliding in the straight section of the recess in the operating slide. The cam surface in this recess then comes in contact with the operating lug of the bolt and cams it upward rotating the bolt counter-clockwise and disengaging the locking lugs on the bolt from their recesses in the receiver. The delay between the initial movement of the operating slide and the unlocking of the bolt enables the bullet to leave the muzzle before unlocking begins, thus relieving the pressure in the barrel before the bolt is opened. The rotation of the bolt also cams the hammer back from the base of the firing pin and withdraws the point of the firing pin into the bolt.

(4) As the operating slide continues its movement to the rear, it carries with it the bolt which slides along the receiver. The empty cartridge case is withdrawn from the chamber

by the extractor (fig. 11). When the mouth of the empty cartridge case clears the breech, the ejector, which is continually pressing on the base of the cartridge, ejects the empty case to the right front through the action of the ejector spring. While the bolt is moving rearward, its rear end bears against the hammer and forces it back and down, thus compressing the hammer spring. The bolt finally loses its rearward motion near the rear end of the receiver. With the bolt at its extreme rearward position, the magazine is uncovered. The magazine follower, actuated by the magazine spring, forces the cartridges upward in the magazine so that the top cartridge lies in the path of the bolt. The rearward movement of the operating slide terminates when

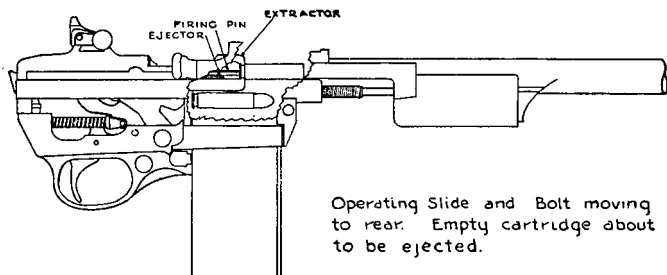


FIGURE 11.—Operating slide and bolt moving to rear.

the rear end of the sleeve section contacts the front face of the receiver.

b. Forward movement.—(1) As the bolt moves forward, actuated by the compressed operating slide spring, the lower front face of the bolt comes in contact with the base of the top cartridge and slides it forward into the chamber (see fig. 12). The hammer, under pressure from the hammer spring, rides on the bottom of the bolt and tends to follow it, but is caught and held by the front end of the sear, which engages in the rear notch on the hammer. When pressure on the trigger is released, the hammer spring imparts a slight forward motion to the hammer, which pivots just enough to push the sear slightly backward and cause its rear end to ride over the trigger lip, completing the cocking action. If,

however, the pressure on the trigger has *not* been released, that is, if the trigger is held back after firing, the rear end of the sear cannot rise above the trigger lip, and the cocking action cannot be completed.

(2) When the bolt approaches its forward position the rim of the cartridge is engaged by the extractor, and the base of the cartridge forces the ejector into the bolt, thus compressing the ejector spring. The operating lug on the bolt is cammed downward by the rear surface of the cam recess in the operating slide, and in this manner the operating slide rotates the bolt clockwise to engage the locking lugs in the receiver. This action locks the bolt. The operating slide then continues to move forward for about $\frac{5}{16}$ inch until the

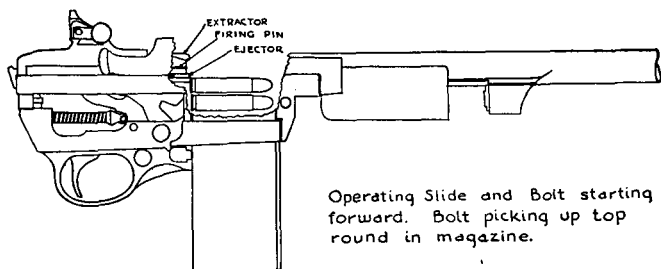


FIGURE 12.—Start of forward movement.

rear end of the straight section of the recess in the operating slide contacts the operating lug on the bolt. The carbine is now ready to be fired again.

(3) The cycle described in (1) and (2) above is repeated each time the trigger is squeezed until the magazine is empty.

SECTION V

OPERATION

■ 21. WHEN TAKEN UP.—The operation of the carbine will be taken up at any convenient time after instruction in care and cleaning has been completed.

■ 22. TO LOAD MAGAZINE.—Insert 15 rounds in the magazine so that the base of each cartridge is close to the rear wall of the magazine and all cartridges are properly alined.

■ 23. TO LOAD CARBINE.—The operation of loading is performed with the piece locked, that is, with the safety of the piece pushed to the right, except in sustained firing. Hold the carbine with the left hand just in front of the magazine opening. With the right hand insert a fully loaded magazine in the magazine opening, making sure that it snaps into place. With the forefinger of the right hand, pull the operating slide smartly all the way back and release it, closing the bolt. The closing of the bolt may be assisted by a push forward on the operating slide handle with the heel of the right hand.

■ 24. TO UNLOAD CARBINE.—The operation of unloading is also performed with the piece locked. Hold the carbine at the small of the stock with the right hand. Press the magazine lock to the left with the forefinger of the right hand, and at the same time withdraw the magazine from the receiver with the left hand, taking care that it does not drop on a hard surface and suffer damage. With the forefinger of the right hand, pull the operating slide all the way back noting that a cartridge or empty cartridge case is ejected. If nothing is ejected, glance into the chamber to see that it is empty. Ease the operating slide forward gently. *CAUTION: Note that removal of magazine does not unload carbine.*

■ 25. TO OPERATE CARBINE AS SINGLE LOADER.—Making sure the receiver is empty, pull the operating slide to the rear and press down on the operating slide catch with the right thumb, locking the operating slide in its rear position. With the thumb and forefinger of the right hand, insert a single cartridge into the chamber. Pull back on the operating slide handle and release it allowing the bolt to go forward.

■ 26. TO FIRE CARBINE.—With the right thumb push the safety from right to left into the off position. Squeeze the trigger for each shot, releasing the pressure on the trigger between shots.

■ 27. **TO SET CARBINE AT SAFE.**—The loaded carbine will be kept locked until the moment for firing. To lock the carbine push the safety from left to right. In this position the trigger cannot be pulled as the safety locks the sear into the sear notch on the hammer. The carbine may be loaded and operated by hand when locked but cannot be fired. To unlock the carbine, push the safety from right to left with the right thumb.

■ 28. **SAFETY PRECAUTIONS.**—The soldier must be impressed with the fact that while any cartridges remain in the receiver after a round has been fired, the carbine is ready to fire. The gun is safe only when it is cleared; in other words, the gun is *never safe* when the bolt is closed.

■ 29. **TO CLEAR CARBINE.**—*a.* To clear the carbine, unload it as in paragraph 24. Engage the operating slide catch in its indent in the receiver. Glance into the chamber to see that it is empty. *Leave the bolt open.*

b. In range firing, execute clear carbine whenever firing ceases.

SECTION VI

IMMEDIATE ACTION AND STOPPAGES

■ 30. **WHEN TAKEN UP.**—Instruction in immediate action and stoppages will be completed before any firing is done by the individual.

■ 31. **IMMEDIATE ACTION.**—*a. General.*—Immediate action is the unhesitating application of a probable remedy for a stoppage. It deals with methods of reducing stoppages and not causes. It is taught as an unhesitating manual operation applied to reduce stoppages without detailed consideration of their causes.

b. Procedure.—(1) *Carbine fails to fire.*—With the right hand palm up, use the little finger to pull the operating slide to the rear. Release the operating rod, and if the operating slide goes fully home, aim and fire. To avoid injury in case of hangfire, the hand is so held that no part of the palm or wrist can be struck by the operating slide in its rapid rearward movement.

(2) *Bolt cannot be locked.*—If after following the procedure prescribed in (1) above, the bolt does not go completely forward and lock, again pull operating handle to the rear. Check for a battered round, dirt or obstruction on the face of the bolt, in the chamber, or in the locking recess. Discard the battered round; remove the obstruction. Release the operating slide handle, aim, and fire.

(3) *Bolt locks but carbine again fails to fire.*—If after procedure prescribed in (1) above, the bolt locks and the carbine still does not fire, again pull operating handle to the rear. If no cartridge is ejected, reduce obstruction in magazine by pressing and rotating the upper cartridges. Release the operating slide handle, aim, and fire.

(4) *Carbine fails to feed.*—Keep carbine in action by manually working operating slide. A detailed examination for the malfunction may be made later when time permits.

c. Detailed examination.—The above procedure of immediate action will almost invariably keep the carbine in action. In case this immediate action is not successful, a more detailed examination for the possible malfunctions listed in paragraph 32e may be made as circumstances permit.

■ 32. STOPPAGES.—*a. General.*—While immediate action and stoppages are closely related as to subject matter, the former is treated separately to emphasize its importance as an automatic and definite procedure to be applied to overcome stoppages. Proper care of the carbine before, during, and after firing will usually eliminate stoppages. Stoppages which cannot be remedied by the application of immediate action can best be eliminated if the soldier has an understanding of the functioning of the weapon and the causes of stoppages.

b. Failure to fire.—(1) *Causes.*—Failure to fire is generally caused by.—

(a) Defective ammunition.

(b) Defective firing pin,

(c) Bolt not fully closed when hammer strikes firing pin.

(2) *Action.*—If the primer of a round is deeply indented, the round is defective. Discard the round. If the primer is not indented or is lightly indented, the firing pin may be short or broken, or the bolt may not have been fully closed.

Check for dirt or some obstruction which does not permit the bolt to lock fully. Remove the obstruction. If the carbine is clean and lubricated, check the firing pin. Replace it if defective. As the disassembly of the bolt requires special tools, replacement of the firing pin should be made by specialists or ordnance personnel.

c. Failure to feed.—(1) *Types.*—Failure to feed is caused by the bolt not going far enough to the rear to pick up a new round. A failure to feed may have a number of causes and results generally in one of the following types of stoppages:

(a) That in which the bolt fails to go fully home.

(b) That in which the bolt does go fully home.

(2) *Action to reduce stoppage of these types.*—Stoppages of the first type may be caused by a battered round, dirt in the locking recesses, an obstruction on the face of the bolt, a dirty chamber, or a ruptured cartridge case, part of which remains in the chamber. Remove the battered round, dirt, or other obstruction; clean the chamber; or remove the ruptured cartridge case. Sometimes this stoppage may be caused by a magazine which has lost its spring tension and does not hold the cartridge firmly in line. Occasionally, when a stoppage of the second type occurs, the spent case is not ejected but is re-fed into the chamber. This condition is caused by lack of lubrication, excessive friction of the moving parts, or lack of sufficient gas pressure due to formation of carbon in the gas port. In any event the bolt has not moved far enough to the rear to permit proper functioning.

d. Failure to extract.—(1) *Causes.*—Failures to extract are generally caused by—

(a) Extremely dirty chamber.

(b) Extremely dirty ammunition.

(c) Improper assembly of the carbine, that is, failure to replace the extractor plunger and spring.

(d) Cartridge case chambered in a hot barrel.

(e) Broken extractor.

(2) *Action.*—(a) When a failure to extract occurs, the bolt may be found fully locked with a spent case in the chamber. Generally, most failures to extract can be remedied by pushing the operating slide fully forward and then pulling

it smartly to the rear. If this does not remove the case, use the cleaning rod.

(b) Sometimes the empty case will be left in the chamber, the extractor ripping through the base of the cartridge. When this occurs the bolt generally will attempt to feed a fresh cartridge into the chamber. It will then be necessary to remove this round before the spent case can be removed.

(c) When the chamber or ammunition is dirty, clean the chamber and discard or wipe off the ammunition. Faulty assembly or a broken extractor will cause recurring failures to extract. Replace missing or broken parts.

e. Table of stoppages.—The following table may be found of value. It includes the stoppages outlined above and others which have not been covered but which may occur occasionally.

TABLE OF STOPPAGES

Malfunction	Cause	Correction by the soldier or ordnance personnel
Failure to extract.....	Dirty or rough chamber.....	Clean chamber.
Failure to feed.....	(1) Dirty or rough chamber.....	(1) Clean chamber.
	(2) Restricted gas port.....	(2) Clean gas port.
	(3) Dirty or improperly lubricated carbine.	(3) Clean carbine and lubricate.
	(4) Damaged magazine.....	(4) Replace magazine.
	(5) Ruptured cartridge case in chamber.	(5) Remove ruptured cartridge case.
	(6) Magazine not fully home.	(6) Push magazine home.
Fires automatically....	Sear broken or remains in open position.	Replace trigger group assembly or sear.
Safety releases when pressure is applied on trigger.	Round heel on safety, or broken safety.	Replace safety.
Pull on trigger does not release hammer.	Deformed hammer or trigger, or worn trigger pin.	Replace defective part.
Hammer releases but gun does not fire.	(1) Bolt not all the way seated.	(1) Clean and lubricate.
	(2) Defective ammunition.	(2) Discard round.
	(3) Broken firing pin.	(3) Replace.

f. Other stoppages.—In the event of stoppages that are not mentioned above and cannot be reduced, the carbine should be turned in for examination and repair.

SECTION VII

SPARE PARTS AND ACCESSORIES

■ 33. SPARE PARTS.—*a.* The parts of any carbine will in time become unserviceable through breakage or wear resulting from continuous usage. For this reason spare parts are supplied. These are extra parts provided with the carbine for replacement of the parts most likely to fail, for use in making minor repairs, and for general care of the carbine. They should be kept clean and lightly oiled to prevent rust. Sets of spare parts should be kept complete at all times. Whenever a spare part is used to replace a defective part in the rifle, the defective part should be repaired or a new one substituted in the spare part set as soon as possible. Parts that are carried complete should at all times be correctly assembled and ready for immediate insertion in the carbine. The allowance of spare parts for the carbine is prescribed in SNL B-28.

b. With the exception of replacements with the spare parts referred to in *a* above, repairs or alterations to the carbine by using organizations are prohibited.

■ 34. ACCESSORIES.—*a. General.*—Accessories include the tools required for assembling and disassembling and for the cleaning of the carbine, the gun sling, spare parts containers, covers, arm locker, etc. Accessories should not be used for purposes other than those for which they are intended. When not in use they should be stored in the places or receptacles provided for them.

b. Arm locker and rack.—The arm locker and the arm rack are used to store or stack carbines to prevent mishandling or pilfering.

c. Cleaning rod and cleaning brush.—The cleaning rod has a handle at one end and is threaded at the other end to receive the patch section or the brush. This rod is of the correct length to prevent damaging the face of the bolt. The cleaning brush is used to clean the bore of the carbine.

d. Gun sling.—The gun sling, fastened to the swivels of the carbine, is adjusted to suit the particular soldier using it. The sling consists of a long web strap, which may be lengthened or shortened as desired. The sling is furnished

to provide the soldier with a means of carrying the carbine. It is not needed in firing.

e. Ruptured-cartridge extractor.—The ruptured-cartridge extractor has the general form of a caliber .30 carbine cartridge. It consists of three parts: the spindle; the head; and the sleeve. The ruptured-cartridge extractor is inserted through the ruptured opening of the case and pushed forward into the chamber. The bolt is let forward without excessive shock so that the extractor of the carbine engages the head of the ruptured-cartridge extractor. As the operating slide is drawn back, the ruptured-cartridge extractor, holding the cartridge on its sleeve, is extracted.

f. Oiler.—The oiler consists of a tube to hold lubricating oil, a steel ring or cap seat and a threaded cap in which is positioned a rod or dropper. The oiler also acts as an anchor pin for the sling.

g. Cleaning rod case.—This fabric container is sewed in five pockets to hold the cleaning brush and the four sections of the cleaning rod.

SECTION VIII

AMMUNITION

■ 35. LOT NUMBER.—When ammunition is manufactured, an ammunition lot number is assigned which becomes an essential part of the marking in accordance with specifications. This lot number is marked on all packing containers and the identification card inclosed in each packing box. It is required for all purposes of record, including grading and use, reports on condition, functioning, and accidents in which the ammunition might be involved. Only those lots of grades appropriate for the weapon will be fired. Since it is impracticable to mark the ammunition lot number on each cartridge, every effort should be made to maintain the ammunition lot number with the cartridges once the cartridges are removed from the original packing. Cartridges which have been removed from the original packing and for which the ammunition lot number has been lost are placed in grade 3. It is, therefore, obvious that when cartridges are removed from their original packings they should be so marked that the ammunition lot number is preserved.

■ 36. GRADE.—Current grades of all existing lots of small arms ammunition are established by the Chief of Ordnance and are published in ordnance field service bulletins. No lot other than one approved by the Ordnance Department will be fired.

■ 37. CARE, HANDLING, AND PRESERVATION.—*a.* Small arms ammunition is not dangerous to handle. Care, however, must be taken not to break or damage the boxes. All broken boxes must be immediately repaired, and all markings transferred to the new parts of the box.

b. The ammunition comes in boxes of 2,000 rounds, packed in 40 cartons of 50 rounds each. Ammunition boxes should

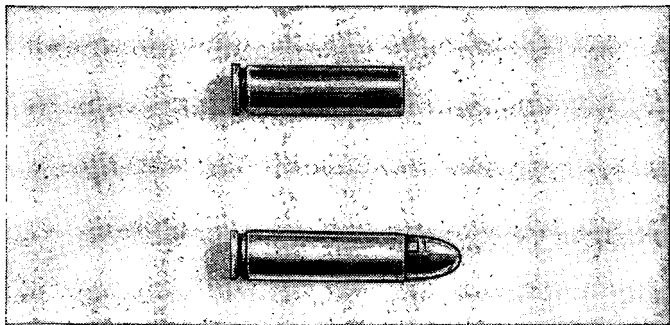


FIGURE 13.—Cartridge and cartridge case, carbine, caliber .30, M1.

not be opened until the ammunition is needed. Ammunition removed from containers, particularly in damp climates, may corrode and thereby become unserviceable.

c. The ammunition should be protected from mud, sand, dirt, and water. If it gets wet or dirty, wipe it off at once. Light corrosion should be wiped off. Cartridges should not be polished to make them look brighter.

d. No carbine ammunition will be fired until it has been positively identified by ammunition lot number and grade.

■ 38. CARTRIDGE, CARBINE, CALIBER .30, M1.—The approximate maximum range for the cartridge is 2,000 yards. Its muzzle velocity is 1,900 feet per second. It is illustrated in figure 13.

SECTION IX

INDIVIDUAL SAFETY PRECAUTIONS

■ 39. RULES.—*a.* Consider every carbine to be loaded until it has been examined and proved to be unloaded. Never trust your memory as to its condition in this respect.

b. Never point the carbine at anyone you do not intend to shoot, nor in a direction where an accidental discharge may do harm.

c. Always unload the carbine if it is to be left where someone else may handle it.

d. Always point the carbine up when snapping the trigger after examination.

e. If it is desired to carry the piece cocked with a cartridge in the chamber, the bolt mechanism will be secured by turning the safety lock to the right.

f. Under no circumstances should the firing pin be let down by hand on a cartridge in the chamber.

g. Never fire a carbine with any grease, cleaning patch, dust, dirt, mud, snow, or other obstruction in the bore. To do so may burst the barrel.

h. Never grease or oil the ammunition or the walls of the carbine chamber. This creates a hazardous pressure on the carbine bolt.

i. See that the ammunition is clean and dry. Examine all live and dummy ammunition. Turn in all cartridges with loose bullets or which appear to be otherwise defective.

j. Do not allow the ammunition to be exposed to the direct rays of the sun for any length of time. This creates hazardous chamber pressures.